

Lab Task 5 :Working with LINQ in C# using Arrays and Collections

Objective:

To practice LINQ operations using both query syntax and method syntax on arrays and generic collections such as List<T>, with a custom class Employee.

Create a class named Employee with the following properties:

```
public class Employee
{
    public string Name { get; set; }

    public double Salary { get; set; }

    public string Department { get; set; }

    public bool IsResigned { get; set; }

    public int Experience { get; set; }
}
```

Given a list/array of employees, perform the following operations using both query syntax and method syntax:

Instructions:

Tasks:

1. **Filter active employees**

Retrieve only the employees who are **currently working** (i.e., `IsResigned == false`).

2. **Sort by salary descending**

Return the list of employees **sorted in descending order of salary**.

3. **Get distinct departments**

Extract a list of **unique department names** from the list.

4. **Check if any employee has zero experience**

Return true if **any employee** has `Experience == 0`.

5. **Project only Name and Salary**

Create a new list of objects that contains **only the Name and Salary** of each employee.

Lab Task 6: C# File I/O – Note Management System

Objective:

Design and implement a C# console application named NoteManagerApp that allows the user to manage personal text-based notes. The application should store all notes inside a specific directory on the D: drive and provide options to create, read, list, copy, move, and delete notes.

1. **SetupNoteDirectory()**
 - a. Create a folder named D:\NotesApp if it does not already exist.
2. **CreateNote(string title, string content)**
 - a. Create a text file named <title>.txt inside NotesApp.
 - b. If it exists, prompt for overwrite.
3. **ReadNote(string title)**
 - a. Display the full contents of the note using File.ReadAllText.
4. **ListAllNotes()**
 - a. List all note titles (i.e., file names without extensions) inside NotesApp.
5. **CopyNote(string originalTitle, string copyTitle)**
 - a. Copy the contents of one note to a new file using File.Copy.
6. **MoveNote(string oldTitle, string newTitle)**
 - a. Rename or move a note using File.Move.
7. **DeleteNote(string title)**
 - a. Delete a note file using File.Delete.
 - b. Confirm with the user before deletion.
8. **ReadNoteLineByLine(string title)**
 - a. Display each line of the note with line numbers using File.ReadLines.